

COMPACT-RANGE VIETORIS POWERS OF COUNTABLE ORDINALS

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ABSTRACT. For every ordinal $\omega < \alpha < \omega_1$, the ordered compact-set space $K(\alpha, \text{ord})$ of Caruvana and Holshouser is second-countable and Lindelöf, but neither σ -compact nor Menger. This answers Question 2.18 of their published paper. We first prove that a second-countable ground space has a second-countable compact-range Vietoris power at countable index. A delayed diagonal then shows that the closed family of surjections onto $\omega + 1$ with zeroth coordinate ω is not σ -compact. Closed inclusions induce closed embeddings at a fixed index, so this obstruction transfers to every ordinal in the stated interval. Explicit coordinate covers yield the non-Menger conclusion. The proofs are self-contained, consolidate the earlier convergent-sequence case with its generalization, and are accompanied by a Lean 4 formalization of the principal ordinal theorem.

Key words and phrases. Vietoris power; ordered compact sets; countable ordinal; Lindelöf space; σ -compactness; Menger property; formal verification.

1. INTRODUCTION

Caruvana and Holshouser [1, Question 2.18, p. 16] ask whether $K(\alpha, \text{ord})$ is σ -compact or Lindelöf for $\omega < \alpha < \omega_1$, explicitly singling out $\alpha = \omega + 1$. This is Question 1 in the earlier arXiv version. All numbered citations to [1] below refer to the published version in Applied General Topology.

Almeida's earlier manuscript [2] treats $\omega + 1$ using finite occurrence bounds and a delayed diagonal. We consolidate that argument and its extension here, giving complete proofs in a single paper. No result from [2] is needed as an unproved input.

Theorem 1.1. For every ordinal $\omega < \alpha < \omega_1$, the space $K(\alpha, \text{ord})$ is second-countable, hence Lindelöf, and is neither σ -compact nor Menger.

Section 2 proves the general countable-base theorem. Section 3 constructs a closed non- σ -compact subspace in the $\omega + 1$ case. Section 4 proves the closed-subspace transfer, which makes the extension to the full interval immediate. Section 5 uses the same diagonal to defeat finite selections from explicit open covers. Both successor and limit ordinals are included; compactness of α itself is unnecessary.

2. THE SPACE AND ITS COUNTABLE BASE

All ordinals carry their order topology, as in [1, Section 1.1, p. 2]. The notation $K(X, \text{ord}, \kappa)$ below denotes the source paper's double-struck K notation for the compact-range subspace of its Vietoris power. For a space X and an infinite cardinal κ , the Vietoris-power topology has basic sets [1, Definition 1.8 and Proposition 1.9, pp. 5-6]

$$[U; F, V] = \{f \in X^\kappa : \text{img}(f) \subseteq U, f(i) \in V_i \ (i \in F)\},$$

where U and all V_i are open and $F \subseteq \kappa$ is finite. Definition 1.14 [1, p. 7] gives the compact-range subspace

$$K(X, \text{ord}, \kappa) = \{f : \kappa \rightarrow X : \text{img}(f) \text{ is compact}\}.$$

The default index for $K(X, \text{ord})$ is the supremum of the cardinalities of the nonempty compact subsets of X , plus ω in cardinal arithmetic. For a countable ordinal α , every such subset is countable, so this index is ω . Accordingly, write

$$K_\alpha = K(\alpha, \text{ord}, \omega) = K(\alpha, \text{ord}).$$

The functions need not be continuous or order-preserving. Compactness refers to the actual image. In a compact-range power at index ω , write

$$T(U) = \{f \in K(X, \text{ord}, \omega) : \text{img}(f) \subseteq U\},$$

$$E(k, U) = \{f \in K(X, \text{ord}, \omega) : f(k) \in U\}.$$

For open U , these are open sets. Basic neighborhoods are tubes intersected with finitely many coordinate restrictions. In particular, every coordinate evaluation $f \mapsto f(k)$ is continuous.

Lemma 2.1. If X is second-countable, then $K(X, \text{ord}, \omega)$ is second-countable.

Proof. Let B be a countable base for X , and let $B^\cup$ be the countable family of finite unions of its members. Consider the basic sets with range bound $W \in B^\cup$, finite coordinate set $F \subseteq \omega$, and coordinate bounds $W_i \in B$ for $i \in F$, all restricted to $K(X, \text{ord}, \omega)$. There are countably many such choices.

Given a compact-range map $f \in [U; F, V]$, cover $\text{img}(f)$ by members of B contained in U . Compactness gives finitely many such members with union W , so $\text{img}(f) \subseteq W \subseteq U$. For each $i \in F$, choose $W_i \in B$ with $f(i) \in W_i \subseteq V_i$. Then

$$f \in [W; F, (W_i)_{i \in F}] \subseteq [U; F, V].$$

The indicated countable family is therefore a base. $\square$

Corollary 2.2. For every countable ordinal α , the space K_α is second-countable and Lindelöf.

Proof. The interval and ray base of a countable ordinal has only countably many possible endpoints. Apply Lemma 2.1 and the implication from second countability to Lindelöfness. $\square$

The compact-image hypothesis is used in the finite-union refinement. No corresponding assertion about the full Vietoris power $V(X^\omega)$ is made here.

3. THE CLOSED OBSTRUCTION IN THE CONVERGENT-SEQUENCE CASE

In this section let $K = K(\omega + 1, \text{ord})$. The ordinal $\omega + 1 = [0, \omega]$ is compact, its finite points are isolated, and every singleton is closed. Define

$$S = \{f \in K : f(0) = \omega, \text{img}(f) = \omega + 1\}.$$

Lemma 3.1. The set S is closed in K .

Proof. The condition $f(0) = \omega$ is closed by continuity of evaluation. For each $n \in \omega$, the condition $n \in \text{img}(f)$ is closed, because its complement is the open tube $T((\omega + 1) \setminus \{n\})$. Intersect these closed conditions. The value ω is already attained at coordinate zero, so the intersection is exactly S . $\square$

Lemma 3.2. If $C \subseteq S$ is compact and $n \in \omega$, then there is $b \in \omega$ such that every $f \in C$ takes the value n at some coordinate $k \leq b$.

Proof. The open sets $E(k, \{n\})$, indexed by $k \in \omega$, cover C , since every member of S is onto $\omega + 1$. Take a finite subcover and let b be the maximum of its coordinate indices, using $b = 0$ for the empty subcover. $\square$

Proposition 3.3. The closed subspace S is not σ -compact. Consequently, $K(\omega + 1, \text{ord})$ is not σ -compact.

Proof. Suppose $S = \bigcup_{n \in \omega} C_n$ with every C_n compact. Apply Lemma 3.2 to C_n and the value n , obtaining a bound $b(n)$. Choose distinct positive integers $j(n) > b(n)$, for example by the recursion

$$j(0) = b(0) + 1, j(n + 1) = \max\{j(n), b(n + 1)\} + 1.$$

Define $g : \omega \rightarrow \omega + 1$ by

$$g(j(n)) = n \ (n \in \omega), g(k) = \omega \ (k \notin \{j(n) : n \in \omega\}).$$

Then $g(0) = \omega$ and $\text{img}(g) = \omega + 1$, so $g \in S$. For each n , its only occurrence of the value n is at $j(n) > b(n)$; thus $g \notin C_n$. This contradicts the cover and proves that S is not σ -compact. Since S is closed, a countable compact cover of K would restrict to one of S . Hence K is not σ -compact either. $\square$

This gives the earlier convergent-sequence obstruction of Almeida [2] in a form ready for the closed-subspace transfer.

4. CLOSED EMBEDDINGS AND THE FULL ORDINAL INTERVAL

Proposition 4.1 (closed-subspace transfer). If A is closed in X , the inclusion induces a homeomorphism of $K(A, \text{ord}, \omega)$ onto the closed subspace $Z_A = \{f \in K(X, \text{ord}, \omega) : \text{img}(f) \subseteq A\}$.

Proof. A subset of A is compact in the subspace topology exactly when it is compact as a subset of X . Thus coordinatewise inclusion identifies the carriers. The set Z_A is closed, being the intersection over $k \in \omega$ of the closed conditions $f(k) \in A$; compare [1, Proposition 1.12].

An ambient basic set with open bounds U and V_i restricts to the basic set with bounds $U \cap A$ and $V_i \cap A$. Conversely, every open bound in A is the intersection of A with an open subset of X . Therefore the two induced topologies agree. $\square$

Corollary 4.2. If A is closed in X and $K(A, \text{ord}, \omega)$ is not σ -compact, then $K(X, \text{ord}, \omega)$ is not σ -compact.

Proof. Closed subspaces of σ -compact spaces are σ -compact: intersect a countable compact cover with the closed subspace. Apply Proposition 4.1 and take the contrapositive. $\square$

Corollary 4.3. For every $\omega < \alpha < \omega_1$, the space $K(\alpha, \text{ord})$ is not σ -compact.

Proof. The initial segment $A = [0, \omega]$ is closed in α , since its complement is an open final segment, and is homeomorphic to $\omega + 1$. The default indexing cardinals for both A and α are ω . Proposition 4.1 therefore identifies $K(\omega + 1, \text{ord})$ with a closed subspace of $K(\alpha, \text{ord})$. Proposition 3.3 and Corollary 4.2 give the conclusion. $\square$

The shared index is essential to this application. The proposition compares powers at a fixed index, rather than asserting preservation of closed inclusions for arbitrary default indices. Corollaries 2.2 and 4.3 answer both parts of [1, Question 2.18].

5. THE Menger STRENGTHENING AND THE COORDINATE BOUND

Recall that a space is Menger if it satisfies $S_{fin}(O, O)$ in the notation of [1, Definition 1.2 and p. 4]: for every sequence of open covers, finite selections from the respective covers can be made whose union covers the space.

Theorem 5.1. For every $\omega < \alpha < \omega_1$, the space K_α is not Menger.

Proof. Put $A = [0, \omega] \subseteq \alpha$ and let

$$S_\alpha = \{f \in K_\alpha : f(0) = \omega, \text{img}(f) = A\}.$$

This is the image of S from Section 3 under the closed embedding of Proposition 4.1, so it is closed in K_α . Here $E(k, \{n\})$ denotes the coordinate set in K_α . For each $n, k \in \omega$, put

$$O_{n,k} = (K_\alpha \setminus S_\alpha) \cup E(k, \{n\}), U_n = \{O_{n,k} : k \in \omega\}.$$

Each $O_{n,k}$ is open because S_α is closed and the finite ordinal n is isolated in α . The family U_n covers K_α : points outside S_α belong to every member, and points in S_α take the value n at some coordinate.

Let $V_n \subseteq U_n$ be arbitrary finite selections. For each selected open set choose a representing coordinate index k such that the set equals $O_{n,k}$. Let I_n be the finite set of these chosen coordinate indices, and define

$$b(n) = \max(\{0\} \cup I_n).$$

Thus the maximum is taken over the coordinate indices k of the selected sets $E(k, \{n\})$. For an empty selection, $I_n = \emptyset$ and $b(n) = 0$.

Use Proposition 3.3's diagonal construction with these bounds, viewing the resulting map as $g : \omega \rightarrow \alpha$. Its image is the compact set A , and $g \in S_\alpha$. For every n and every $k \in I_n$, we have $k \leq b(n) < j(n)$, whereas the only coordinate at which g takes the value n is $j(n)$. Hence $g \notin E(k, \{n\})$ and $g \notin O_{n,k}$. It belongs to none of the selected open sets, so the selections cannot cover K_α . $\square$

Corollaries 2.2 and 4.3 and Theorem 5.1 prove Theorem 1.1. The same compact image $[0, \omega]$ works throughout the interval, including noncompact limit ordinals.

6. CLASSIFICATION AND RELATION TO THE PUBLISHED RESULTS

Together with the discrete cases and [1, Proposition 2.17], the result gives the following classification under the default indexing convention:

Ordinal α	Lindelöf	σ -compact	Menger
$0 \leq \alpha \leq \omega$	Yes	Yes	Yes
$\omega < \alpha < \omega_1$	Yes	No	No
$\alpha \geq \omega_1$	No	No	No

For $0 < \alpha \leq \omega$, the ground space is countable and discrete, and compact images are finite. For a nonempty finite word s in α , consider the maps extending s whose values lie in $\text{img}(s)$. Their ranges are forced to equal $\text{img}(s)$, so the induced topology is just the product topology on the remaining coordinates, with finite discrete factor $\text{img}(s)$. Each such subspace is compact, as in [1, Remark 1.13, p. 7, and Lemma 2.14, p. 13]. Every finite-range map belongs to one of them, and there are countably many words s . Thus K_α is σ -compact and hence Menger and Lindelöf. The case $\alpha = 0$ is empty.

The endpoint $\alpha = \omega$ agrees with [1, Theorem 2.16(d), p. 14]. For uncountable ordinals, [1, Proposition 2.17, p. 15] proves failure of Lindelöfness, which also excludes σ -compactness and the Menger property.

Theorem 2.19 of [1] concerns the full power $V(\omega^\omega)$, including functions with noncompact image, and so does not conflict with Lemma 2.1. Example 2.35 of [1] concerns $K(\mathbb{R}, \text{ord})$ at the default uncountable index, whereas Lemma 2.1 concerns $K(\mathbb{R}, \text{ord}, \omega)$. The continuous range map of [1, Proposition 2.27] goes from the ordered space to the ordinary Vietoris hyperspace; covering properties of its image need not lift to its domain. Our proof uses the defining topology directly and requires no compact-covering-map assertion from [1].

7. FORMAL VERIFICATION AND SOURCE CORRESPONDENCE

The accompanying Lean 4 project uses actual mathlib ordinals. For $a : \text{Ordinal}$, the ground space is the subtype $\{b : \text{Ordinal} : b < a\}$, with the topology inherited from the ordinal order topology. Its convexity identifies this with its own order topology. The carrier consists of maps from the natural numbers whose actual ranges are compact. Its topology is induced from the topology generated by the range and finite-coordinate basic sets in [1, Definition 1.8].

Checked principal declaration: `VietorisOrdinals.ordinal_full`

This theorem quantifies over every $\omega < a < \omega_1$ and proves second countability, Lindelöfness, non- σ -compactness and failure of the standard Menger property. The formalization proves the general countable-base lemma, closedness of the fixed-image family and the occurrence bounds, and constructs the delayed diagonal inside each ordinal. Its delayed positions use the natural-number pairing bijection, as in [2], instead of the recursion in Proposition 3.3. Both constructions give distinct positive positions beyond the prescribed bounds.

For the Menger proof, the Lean code uses the increasing finite unions

$$H_{n,m} = \bigcup_{k \leq m} O_{n,k} = (K_\alpha \setminus S_\alpha) \cup \bigcup_{k \leq m} E(k, \{n\}).$$

For finite selections of these sets, it chooses a representing index m for each selected set and takes the maximum of those finitely many indices, with default zero. Every coordinate index k occurring in the selected $E(k, \{n\})$ terms then satisfies $k \leq m \leq b(n)$. This is the same coordinate bound used in Theorem 5.1. The definition of Menger in the project allows arbitrary open covers; the displayed countable covers suffice to refute it.

The checked proof establishes the principal ordinal theorem directly. Proposition 4.1 and its abstract transfer corollary are proved in this manuscript but are not separately formalized in the supplied project. The uncountable classification is attributed to [1, Proposition 2.17] and is outside this formalization.

The recorded verification of 28 September 2026 used Lean 4.19.0 and mathlib v4.19.0; its transitive axiom audit reports only `propext`, `Classical.choice` and `Quot.sound`. The accompanying record preserves theorem types, compiler output, dependency revisions and source hashes.

The default index is ω by countability, and the cardinal bound on subsets is also formalized. The correspondence between the published definitions and the formal model is documented mathematical interpretation. The original verification files cite the earlier arXiv version; the definitions are unchanged in the published version, where the question is numbered 2.18.

Formalization and verification record: github.com/LucasAlves8Sp/vietoris-countable-ordinals

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AI DISCLOSURE

Almeida proposed and guided the extension. ChatGPT-Astra developed the main proof, Lean formalization, verification work, and manuscript under Almeida's direction.

REFERENCES

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- [2] Lucas Alves de Almeida, On the Vietoris Power of a Convergent Sequence, revised unpublished manuscript, 2026, 3 pp., with accompanying Lean 4 verification. The proof is incorporated in Section 3 of the present paper.